

Experiment 3

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Aim: To create a Cryptocurrency using Python and perform mining in the Blockchain created.

The experiment demonstrates transactions, mining, mining rewards, multiple blockchain nodes, consensus, chain validation, and synchronization of nodes.

Theory

1. Blockchain

A **Blockchain** is a decentralized digital ledger that stores data in a sequence of connected blocks. Each block contains transactions, a timestamp, the previous block's hash, and proof of mining. The blocks are linked using cryptographic hashes, making unauthorized modification difficult.

2. Cryptocurrency

A **cryptocurrency** is a digital currency that can be transferred between participants using blockchain technology. In this experiment, transactions are stored in blocks and a miner receives a cryptocurrency reward after successfully mining a block.

In our implementation, **Siddhant** is used as the miner/reward receiver.

3. Transactions

A transaction contains three main details:

- Sender
- Receiver
- Amount

For example:

Sender: Siddhant

Receiver: Bob

Amount: 200

Transactions are first stored in the pending transaction list. They are added to a block when mining takes place.

4. Mining

Mining is the process of collecting pending transactions and creating a new block by solving a Proof-of-Work puzzle.

When a block is successfully mined, the miner receives a cryptocurrency reward.

In this experiment, the miner is:

Siddhant

5. Proof-of-Work

Proof-of-Work is a consensus mechanism in which the miner repeatedly searches for a suitable proof/nonce that satisfies a cryptographic condition.

The SHA-256 hash is calculated repeatedly until the required condition is satisfied.

6. Mining Reward

After successfully mining a block, the miner receives a reward transaction.

For this experiment:

Sender: 0

Receiver: Siddhant

Amount: 1

This represents the cryptocurrency mining reward.

The PDF describes transactions being added to a mined block along with a reward for the miner.

7. Multi-Node Blockchain

A multi-node blockchain consists of multiple computers or nodes, where each node maintains its own copy of the blockchain.

The experiment uses **three nodes**:

Node 1 → Port 5001

Node 2 → Port 5002

Node 3 → Port 5003

Each node can communicate with the other nodes. The PDF specifically describes a three-node blockchain and communication between peer nodes.

8. Consensus Mechanism

When different nodes have different blockchain lengths, the **Longest Valid Chain Rule** is used.

The process is:

Current Chain



Ask connected nodes for their chains



Compare chain lengths



Check validity



Find longest valid chain



Replace shorter chain

This allows the nodes to maintain a consistent blockchain.

9. Chain Validation

The `is_valid()` function checks whether the blockchain is valid.

It verifies that:

1. The previous hash of each block is correct.
2. The Proof-of-Work condition is satisfied.
3. The blocks are properly connected.

10. Removal of Pending Transactions

After pending transactions are copied into a newly mined block, the pending transaction list is cleared.

This ensures that already-mined transactions are not included again in another block. The PDF specifically demonstrates this modification.

Conclusion

The experiment successfully demonstrated the development of a Python-based cryptocurrency and a multi-node blockchain network. Transactions were added to the pending transaction list and were included in a newly mined block along with a mining reward for Siddhant. Proof-of-Work was used to mine blocks, while cryptographic hashing maintained the integrity of the blockchain.

Three blockchain nodes were created and connected to demonstrate decentralization. The Longest Valid Chain Rule was used as the consensus mechanism to maintain consistency between nodes. Blockchain validation was also performed to verify the integrity of the chain. Pending transactions were removed after being successfully added to a block.